Mini-Project: Singular Value Decomposition

StreamFlix — Building a Recommender That Works

Weight: 100 points (+10 bonus available) **Due:** ____________________ **Work in groups of:** ________

The Scenario

You have been hired as a data scientist at StreamFlix, a regional streaming service with a few hundred thousand subscribers.

StreamFlix has a problem. Cancellations are up 12% over two quarters, and the exit survey shows the same complaint again and again:

"There's never anything to watch."

The catalogue has thousands of films, so the problem is not supply — it is that the homepage shows essentially the **same popular titles to everybody**. A subscriber who loves quiet foreign drama sees the same action blockbusters as everyone else, concludes the service is not for them, and leaves.

Two people want something from you.

- **The VP of Product** wants a system that predicts how a subscriber would rate a film they have not seen, so the homepage can be personalised. She wants to know how much better it is than what StreamFlix does today, which is simply ranking by average rating.
- **The Head of Content Licensing** wants something different and, for him, more valuable. Every year he spends millions renewing licences largely on instinct. He wants to know: **what are the actual dimensions along which our audience divides?** Not genre labels that a studio assigned, but the real fault lines in how *our* subscribers behave. If he knew there were, say, four distinct viewing tastes, he could buy deliberately for each.

The engineering team has exported the ratings log. There is a catch, and it is the whole difficulty of the project:

The ratings matrix is 98.6% empty. There are 610 subscribers and 9,724 films, which is 5,931,640 possible ratings — but only 100,836 were actually given. Almost every cell you would like to know is blank. Your job is to fill them in.

The Data

MovieLens "latest-small" — GroupLens Research, University of Minnesota.

Download: `https://files.grouplens.org/datasets/movielens/ml-latest-small.zip`

Dataset home: `https://grouplens.org/datasets/movielens/`

0.93 MB zipped. The starter notebook downloads and caches it for you.

Two files matter:

File	Rows	Columns
`ratings.csv`	100,836	`userId`, `movieId`, `rating` (0.5–5.0 in half-steps), `timestamp`
`movies.csv`	9,742	`movieId`, `title`, `genres` (pipe-separated)

This is real data from real people, not a teaching toy. Every user in it rated at least 20 films.

Cite the source in your report. MovieLens data is free for research and education, but GroupLens asks to be credited. See the `README.txt` inside the zip for the citation and licence terms — read it.

What You Submit

1. **Your completed notebook** (`.ipynb`), *with all output cells saved.* A notebook that has to be re-run to be read will lose marks. Start from `SVD_MiniProject_Starter.ipynb`.
2. **A one-page business memo** (Part 7) — inside the notebook or as a separate PDF.

Rules.

- Build the model with `numpy` / `pandas` / `matplotlib`. **Do not use a recommender library** (`surprise`, `implicit`) or `sklearn.decomposition`. The point is to build it from the SVD up. You may use `sklearn` only to *check* a result, and must say so.
- **Do not change the random seed or the train/test split** in the starter. Your numbers must be comparable to everyone else's.
- **Never fit on the test set.** Every mean, bias, and factor comes from training data only. Test ratings are touched exactly once, to score.

The Tasks

Part 1 — Know your data — 10 points

Describe what you are working with before you model it: the distribution of rating values, ratings per user and per film (min / median / max), the ten most-rated films, and a histogram of ratings-per-film on a log scale.

Written: what does the shape of this data imply for a recommender? Which users and films will be hardest to predict, and why?

Part 2 — Baselines — 15 points

Build three deliberately dumb predictors — global mean, user mean, item mean — and score each by RMSE on the held-out set.

You cannot claim SVD helped unless you know what it beat. Note that some test films have no training ratings at all; decide what to predict for them and justify it.

Part 3 — A first SVD attempt — 20 points

The obvious approach. SVD needs a complete matrix, so fill the gaps with each film's mean rating, subtract each user's mean, and take the SVD. Sweep k, reconstruct $\boldsymbol{A}_k = \boldsymbol{U}_k \boldsymbol{\Sigma}_k \boldsymbol{V}_k^{\mathsf{T}}$, add the means back, and plot test RMSE against k.

Written, and this carries most of the marks:

1. Compare your best SVD result with your best Part 2 baseline. **Does SVD actually win?**
2. What fraction of the energy $\sum_{i \le k} \sigma_i^2 / \sum_i \sigma_i^2$ does $k = 1$ alone capture? Does the energy criterion pick the same k that the RMSE curve does? **Explain the discrepancy.**

If this approach loses to a baseline, say so. That is a real result, and diagnosing it correctly is worth full marks. Hiding it is not.

Part 4 — Do it properly — 25 points

Most of the signal in a ratings matrix is not taste at all: some users rate generously, and some films are simply better. Model that first, then let SVD work on the remainder.

Step 1. Fit a regularised bias model $b_{ui} = \mu + b_u + b_i$ (the starter gives the formulas; begin with $\lambda = 10$) and score it alone.
Step 2. Form the residual matrix — observed entries become $r_{ui} - b_{ui}$, unobserved become 0 — run the SVD on *that*, and predict $\hat{r}_{ui} = b_{ui} + (\boldsymbol{U}_k \boldsymbol{\Sigma}_k \boldsymbol{V}_k^{\mathsf{T}})_{ui}$.

Report RMSE vs k (include $k = 0$), plot Parts 3 and 4 on the same axes, and give your best k with the percentage improvement over your best baseline.

Written: why is zero-filling the *residuals* legitimate when zero-filling the *ratings* would have been nonsense?

Part 5 — What are the latent factors? — 15 points

This is the Head of Licensing's question. For each of the first three factors, list the films loading most positively and most negatively, with genres, and **invent a plain-English name** for the factor.

Restrict to films with at least 50 ratings and state your threshold — otherwise the extremes are obscure titles rated twice and you will read meaning into noise.

Be honest. Some factors are genuinely uninterpretable. "Factor 3 has no clean reading, and here is the evidence" earns full marks. Inventing a tidy story the data does not support does not.

Part 6 — Actually recommend something — 5 points

For three users: show their top-rated training films, then recommend five they have not seen (excluding films with under 50 ratings).

Written: find at least one recommendation that looks *wrong* and explain what confused the model.

Part 7 — The business memo — 10 points

One page maximum, to the VP of Product. No formulas, no jargon, and no RMSE figure without translating what it means in practice.

Cover: what you built; how much better than the obvious approach, expressed so a product person can feel it; what the factors say about the audience plus one concrete licensing recommendation; two honest limitations; and what you would do with two more weeks.

Stretch goals — up to 10 bonus points

- **Shrink the correction.** Weight the low-rank term by $w \in [0, 1]$ and tune it. Does $w < 1$ beat $w = 1$?
- **An honest split.** The random split lets a user's future predict their past. Re-split by `timestamp`. Does RMSE get worse, and why is the worse number the more trustworthy one?
- **Cold start.** Show what your model does for a brand-new user with zero ratings, then propose a fix.
- **Beyond RMSE.** RMSE judges predicted numbers; recommenders are judged on ranking. Implement precision@10. Does it prefer a different k?
- **Scale.** Time the dense SVD and compare with a truncated solver. At what matrix size does the dense approach stop being practical?

Grading Rubric

Component	Pts	What earns full marks
Data exploration	10	All five items present; the written answer connects sparsity to concrete prediction difficulty rather than restating numbers.
Baselines	15	Three baselines correct, computed from training data only, with the empty-film case handled and justified.
First SVD attempt	20	Correct pipeline and RMSE-vs-k plot; **honest reporting** of whether it beat the baseline; the energy-vs-RMSE discrepancy correctly explained.
Bias + residual SVD	25	Regularised biases correct; residual SVD implemented properly; improvement demonstrated over both the baseline *and* Part 3; the zero-filling question answered convincingly.
Factor interpretation	15	Sensible popularity threshold; both ends of each factor examined; names justified by evidence; uninterpretable factors admitted rather than fabricated.
Recommendations	5	Correct exclusion of already-seen and unpopular films; at least one failure case identified and diagnosed.
Business memo	10	Genuinely non-technical; improvement translated into business terms; limitations honest and specific rather than generic.
Total	**100**	

How to lose marks fast

- **Scoring on the training set.** If your RMSE looks wonderful, suspect this first.
- **Computing means over the full dataset** instead of the training split. Subtle, and it inflates every result.
- **Reporting a number without a baseline.** "RMSE 0.87" means nothing on its own.
- **Interpreting factors from rarely-rated films.** You will produce confident nonsense.
- **Claiming a factor means something** because it sounds plausible, with no supporting evidence.
- **A memo full of jargon.** If your VP would need to ask what a singular value is, rewrite it.

A closing word on what this project is really teaching.

You already computed an SVD by hand on a 3×2 matrix of three people and two movies. The mathematics here is *identical* — the same $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$, the same singular values ordered by importance, the same truncation. Nothing new has been added.

What changed is everything around it: the matrix is almost entirely missing, so what you put in the gaps matters more than the factorisation itself; there is no “right answer” to check against, so you need a held-out set and a baseline; and the output has to mean something to somebody who will never see a matrix.

That gap — between a technique that works on paper and a technique that works on data — is the actual subject of this assignment.

Data citation. F. Maxwell Harper and Joseph A. Konstan (2015). “The MovieLens Datasets: History and Context.” *ACM Transactions on Interactive Intelligent Systems* 5(4), Article 19. https://doi.org/10.1145/2827872